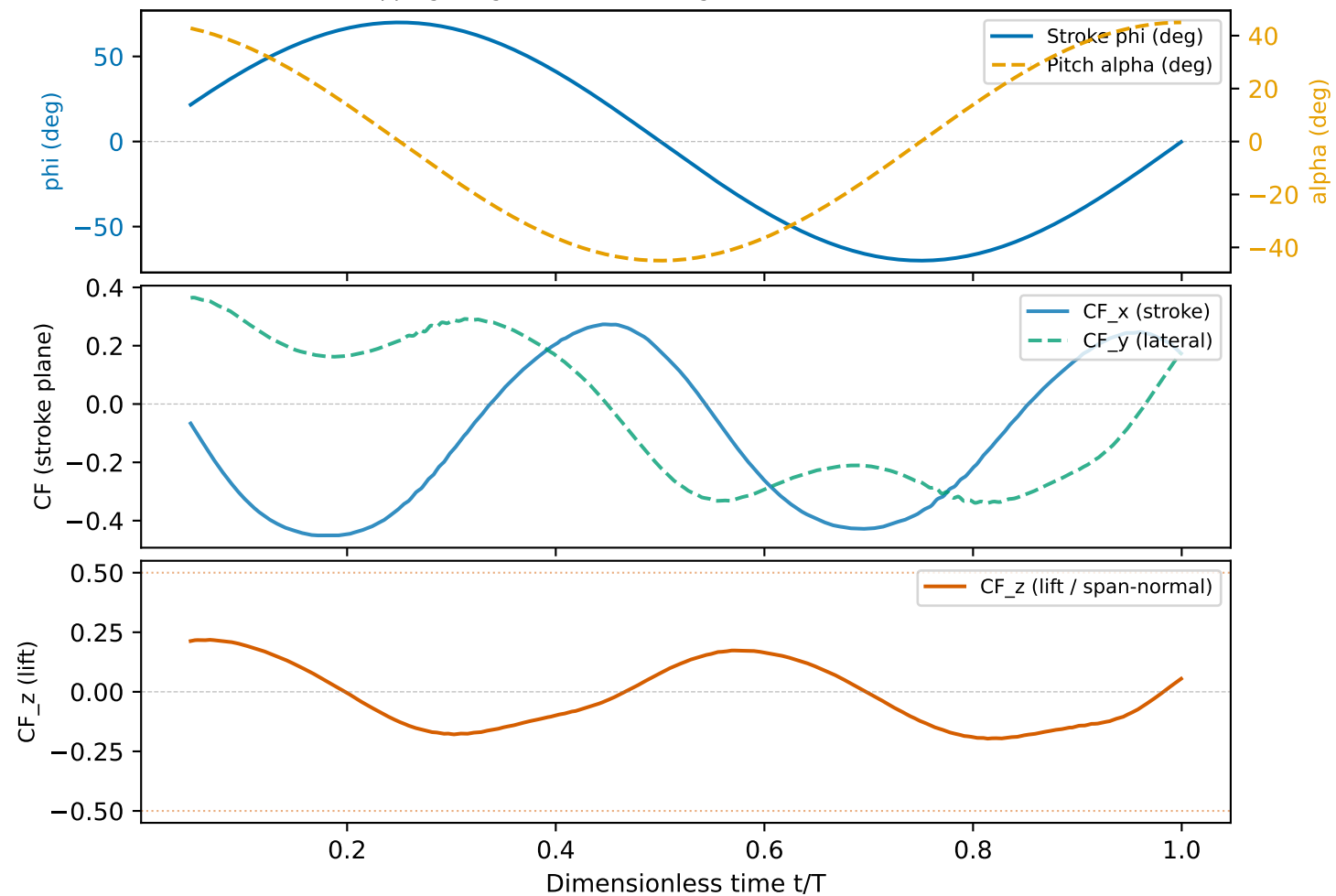


Flapping Wing Forces — coarse grid 64x32x64, $f^*=1$, $Re \sim 100$



Forces normalized by $q_{tip} \times S = 624.8$ ($q_{tip} = 0.5 \rho V_{tip}^2$, $V_{tip} = 23.0$)